# Duc Cuong Vu

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## Educations

**Ph.D. in Electrical and Electronic Engineering** *Aug 2026 - present*
*Supervised by Prof. Lihua Xie* *Nanyang Technological University (NTU), Singapore*

**Master Degree in Control Engineering and Automation** *Jul 2024 - Jul 2026*
*Supervised by Prof. Tung Lam Nguyen* *Hanoi University of Science and Technology (HUST), Hanoi, Vietnam*
*Thesis: Design control structures for parallel platforms in maritime applications*

**Bachelor Degree in Control Engineering and Automation** *Oct 2020 - Apr 2024*
*Supervised by Prof. Tung Lam Nguyen* *Hanoi University of Science and Technology (HUST), Hanoi, Vietnam*
*Thesis: Balancing, Motion Planning, and Tracking Control for Ballbot Systems*

## Work experiences

**R&D Robotics Engineer at VinRobotics, Vingroup** *Sep 2025 - Jul 2026*
*Full-time, supervised by Dr. Minh Nhat Vu* *Hanoi, Vietnam*
*System identification, Kinematics calibration, Advanced control, impedance control, motion planning, optimization, obstacle avoidance, manipulator, humanoid, EtherCAT, ROS2, conduct scientific paper, etc.*

**Research Project Assistant at HUST** *Oct 2021 - Dec 2026*
*Contract (Hybrid), supervised by PhD. Chinh Hoang Duc and Assoc.Prof.PhD. Tung Lam Nguyen* *Hanoi, Vietnam*
*Safety control, Observer, Nonlinear control, Motion control/planning, ball-balancing robot, parallel mechanism, underwater vehicles, vibration compensation, etc.*

## Selected publications

[9] **Vu, D. C.**, & Nguyen, T. L. (2026). Safe-Ballbot: Tilt-Consistent MPC with Control Barrier Functions for Ballbot Navigation. *arXiv preprint arXiv:2611.03307*. [pdf]

[8] **Vu, D. C.**, Nguyen, V. T., Nguyen, D. H, Nguyen, M. C., Tran, V. T., & Vu, M. N.(2026). Real-Time Conformal-Seeded Hybrid Inverse Kinematics for Offset Redundant Manipulators. *IEEE RA-L*. [pdf]

[7] Nguyen, M. C.†, **Vu, D. C.**†, Vu, V. A., Pham, D. T., Nguyen, T., Phung, M. H., Ngo, A. V., & Vu, M. N. (2026). Adaptive Whole-Body MPC with Optimal Posture Planning for Humanoid Robots under Unknown Payloads. *IEEE-RAS 25th International Conference on Humanoid Robots (Humanoids 2026)*. [pdf] †equal contribution

[6] Nguyen, T. T. H., **Vu, D. C.**, Pham, M. D., Nguyen, T. L., & Nguyen, T. V. A. (2026). Unifying hierarchical sliding mode control and control barrier function for tilt angle constraint of a ball-balancing robot. *IET Cyber-Systems and Robotics*. [pdf]

[5] **Vu, D. C.**†, Tran, S.†, Nguyen, T. L., & Hoang, D. C. (2025). Glocal trajectory generation and tracking control for AUVs with optimal coverage sensor networks. *Ocean Engineering*. [pdf] †equal contribution

[4] **Vu, D. C.**, Nguyen, D. H., Vu, M. N., & Nguyen, T. L. (2025). Lagrangian-based modeling and safety-critical controls for Stewart platforms under marine operations. *Ocean Engineering*, 340, 122142. [pdf]

[3] Pham, M. D., **Vu, D. C.**, Nguyen, T. T. H., Nguyen, T. V. A., Vu, M. N., & Nguyen, T. L. (2025). CBFs-based model predictive control for obstacle avoidance with tilt angle limitation for ball-balancing robots. *IEEE Access*. [pdf]

[2] **Vu, D. C.**, Nguyen, D. H., & Nguyen, T. L. (2025). A novel approach of consensus-based finite-time distributed sliding mode control for Stewart platform manipulators motion tracking. *Results in Engineering*. [pdf]

[1] **Vu, D. C.**, Pham, M. D., Nguyen, T. T. H., Nguyen, T. V. A., & Nguyen, T. L. (2024). Time-optimal trajectory generation and observer-based hierarchical sliding mode control for ballbots with system constraints. *International Journal of Robust and Nonlinear Control*. [pdf]

## Honours & awards

**NTU Research Scholarship** *Nanyang Technological University (NTU), Singapore*

**Master, PhD Scholarship Programme** *Vingroup Innovation Foundation (VINIF)*

**Best Thesis Defense Award** *Hanoi University of Science and Technology*

## Academic activities

**Invited review for** *Nonlinear Dynamics, Ocean Engineering, IEEE Access, IEEE Robotics and Automation Letters (RA-L); International Conference on Intelligent Robots and Systems (IROS 2026), IEEE RO-MAN 2026, Humanoids 2026, International Conference on Robotics and Automation (ICRA 2027).*